

Supplementary Material for ANCHORPATH: Defeating High-Sparsity Collapse in Structured Pruning

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Notation and Restated Results

This supplement is self-contained. We first recall the objects the main paper defines and that the analysis below relies on; the numbering here is local to this document.

Objective. Writing $s \in \{0, 1\}^U$ for a structured pruning mask over U removable units and p_s for the pruned model, ANCHORPATH minimizes the deviation from the dense model p_0 measured as calibration KL to the teacher,

$$\mathcal{R}(s) = \mathbb{E}_{x \sim \mathcal{C}} \text{KL}(p_0(\cdot | x) \| p_s(\cdot | x)), \quad (1)$$

a self-distillation objective that needs only forward passes.

One-shot (dense-anchor) quadratic. Expanding (1) once at the dense model gives the one-shot second-order surrogate

$$\mathcal{R}(s) \approx \frac{1}{2} s^\top H s, \quad H_{uv} = \mathbb{E}_x [\delta z_u^\top F^0 \delta z_v], \quad (2)$$

where F^0 is the dense Fisher metric and δz_u is the marginal logit change of removing unit u from the dense model. This is the $K=1$ case; all one-shot baselines instantiate (2).

Anchored surrogate along the path. At operating point s_k with accumulated logit drift $\Delta z^{(k)}$, holding the metric fixed at F^0 but *refreshing* the ablation perturbations $\delta z_u^{(k)}$ at the current model, the implemented anchored surrogate is

$$g_u^{(k)} = \mathbb{E}_x [\Delta z^{(k)\top} F^0 \delta z_u^{(k)}], \quad (3)$$

$$H_{uv}^{(k)} = \mathbb{E}_x [\delta z_u^{(k)\top} F^0 \delta z_v^{(k)}]. \quad (4)$$

The first-order term $g^{(k)}$ is exactly zero at the dense step ($\Delta z^{(0)}=0$) and revives along the path; one-shot discards it. The greedy marginal gain of adding unit u to the current increment set D is

$$\Delta(u | D) = \frac{g_u^{(k)} + \frac{1}{2} H_{uu}^{(k)} + \sum_{v \in D} H_{uv}^{(k)}}{c_u}, \quad (5)$$

with c_u the cost of unit u .

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Algorithm 1 ANCHORPATH (calibration-only anchored continuation)

Input: dense p_0 , calibration $\mathcal{C}$, reachable target cost B , steps K , refresh fraction τ

Output: pruning mask s_K

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1:  $s_0 \leftarrow \mathbf{0}$ ; cache teacher top- $r$  logits and dense ablation
   features on  $\mathcal{C}$ 
2: for  $k = 0$  to  $K - 1$  do
3:    $\mathcal{S} \leftarrow$  surviving units; if  $k=0$  then use cached features
     else re-ablate top- $\tau$  of  $\mathcal{S}$  by staleness
4:   rebuild anchored  $g^{(k)}, H^{(k)}$  via Eqs. (3)–(4);  $D \leftarrow \emptyset$ 
5:   while  $c^\top s_k + \sum_{u \in D} c_u < b_{k+1}$  do
6:      $u^* \leftarrow \arg \min_{u \in \mathcal{S} \setminus D: u \text{ admissible}} \Delta(u | D)$  (Eq. 5)
7:      $D \leftarrow D \cup \{u^*\}$ 
8:   end while
9:    $s_{k+1} \leftarrow s_k \cup D$ ; recompute drift  $\Delta z^{(k+1)}$ 
10: end for
11: return  $s_K$  ( $B \leq c^\top s_K < B + c_{\max}$ )

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Algorithm.

Step-splitting bound. The result the theory section proves is restated here for reference.

Proposition 1 (Step-splitting with surrogate mismatch). *Suppose the exact local Taylor remainder obeys $|\xi_k(\delta_k)| \leq L \|\delta_k\|_{\bar{H}^{(k)}}^3$ and the implemented anchored surrogate $\hat{Q}_k(\delta_k) = g^{(k)\top} \delta_k + \frac{1}{2} \delta_k^\top H^{(k)} \delta_k$ differs from the exact local quadratic $Q_k(\delta_k) = \bar{g}^{(k)\top} \delta_k + \frac{1}{2} \delta_k^\top \bar{H}^{(k)} \delta_k$ by at most ε_k . For a path of total Fisher length $\bar{R}$ split into K equal-length steps,*

$$\left| \mathcal{R}(s_K) - \mathcal{R}(s_0) - \sum_{k=0}^{K-1} \hat{Q}_k(\delta_k) \right| \leq \frac{LR^3}{K^2} + \sum_{k=0}^{K-1} \varepsilon_k.$$

The first term isolates the ideal benefit of step splitting; the second records the mismatch from retaining F^0 instead of the exact $F^{(k)}$ and from partial feature refresh. The proof, the derivation of (2)–(5) from (1), and the $K=1$ equivalence are given in Part B below.

Appendix Overview

This appendix is organized into five self-contained parts, ordered by the questions a reader asks in turn: how the method

runs, why it works, how it is evaluated, what the full results show, and where it applies. Each begins on a fresh page.

Part A — Method and Algorithm Details (App. A–C).

The three-stage calibration-only pipeline, the Fisher–Gram curvature estimator (App. B), the interaction-flagged refresh algorithm (Alg. 2), and the complete offline-cost accounting (App. C).

Part B — Theoretical Analysis (App. D–I). Proof of the $1/K^2$ step-splitting bound (App. D); derivation of the pruning quadratic (App. E) and the revived first-order term (App. F) from the KL risk; the exact $K=1$ equivalence (App. G); one-shot baselines as special cases (App. H); and why the flagged refresh suffices (App. I).

Part C — Experimental Setup (App. J–L). Models and protocol (App. J), baseline configurations (App. K), and metric definitions (App. L).

Part D — Full Results and Analysis (App. M–T).

Per-corpus perplexity across models (App. M); full downstream breakdown (App. N); the *white-box selection analysis*, attention-head pruning, mask overlap, and the attention-budget salvage (App. O); full continuation traces (App. P); the ablation grid (App. Q); the band-placement schedule study (App. R); robustness and sensitivity (App. S); and deployment efficiency (App. T).

Part E — Limitations and Future Work (App. U–V).

The operating range and boundaries of the calibration-only regime (App. U), and future directions that build on the regime above (App. V).

Algorithm 2 Interaction-Flagged Refresh (step $k > 0$)

Input: cached features $\{\Phi_u\}$, last increment D , surviving $\mathcal{S}$, fraction τ

Output: updated $H^{(k)}, g^{(k)}$

- 1: accrue $\sigma_w += \sum_{u \in D} |H_{wu}|$ for $w \in \mathcal{S}$ // cumulative since last refresh
 - 2: $\mathcal{R} \leftarrow$ top- τ of $\mathcal{S}$ by σ ; $\sigma_w \leftarrow 0$, re-ablate each $w \in \mathcal{R}$, update Φ_w
 - 3: recompute base logits z_{s_k} (one forward); $\Phi_\Delta \leftarrow \phi(z_0 - z_{s_k})$
 - 4: $H^{(k)} \leftarrow \frac{1}{P} \Phi \Phi^\top$; $g^{(k)} \leftarrow \frac{1}{P} \Phi \Phi_\Delta^\top$
-

offline cost is $(1 + (K-1)\tau)MF + O(K \cdot F)$, i.e. a small constant multiple of one shot ($4.5\times$ at $K=8, \tau=0.5$). The greedy fill is $O(M^2)$ vector work per step, negligible next to forwards. **Deployment cost is unchanged:** ANCHORPATH outputs a mask, so the pruned model’s parameters, FLOPs, and latency are identical to a one-shot model at the same sparsity; the continuation buys a better mask at the same deployed budget.

Part A: Method and Algorithm Details

A The ANCHORPATH Pipeline

ANCHORPATH has three offline stages, all calibration-only and gradient-free. **(1) Teacher cache.** One forward pass over $\mathcal{C}$ stores, per token, the dense top- r logit indices, probabilities, and logits ($r=256$); these fix a common coordinate system in which every later perturbation is expressed, so all curvatures are anchored to p_0 . **(2) Step-0 ablation.** For every unit u a single masked forward yields the marginal logit perturbation $\delta z_u = z_0 - z_{\{u\}}$ on the teacher coordinates; with layer replay this costs $\approx$ one forward per unit, amortized once. **(3) Continuation.** Algorithm 1 traces the budget path, re-ablating only flagged units at each of the K steps (Alg. 2). Stage (2) is exactly the one-shot pruner’s curvature build, so $K=1$ reuses it verbatim.

B The Fisher–Gram Curvature Estimator

We never materialize the $V \times V$ Fisher. For a token with teacher top- r probabilities $q \in \mathbb{R}^r$ and a logit perturbation $a \in \mathbb{R}^r$ on those coordinates, the Fisher quadratic form is the *centered, probability-weighted* inner product

$$a^\top F^0 a = \sum_i q_i a_i^2 - \left(\sum_i q_i a_i \right)^2 = \|\phi(a)\|_2^2, \quad (6)$$

where $\phi(a)_i = \sqrt{q_i}(a_i - \bar{a})$ and $\bar{a} = \sum_i q_i a_i$ is the probability-weighted mean. Stacking $\phi(\delta z_u)$ over all calibration tokens into a feature matrix $\Phi_u \in \mathbb{R}^{P \times r}$, the curvature is a Gram product $H_{uv} = \frac{1}{P} \langle \Phi_u, \Phi_v \rangle$ and the drift term is $g_u = \frac{1}{P} \langle \Phi_\Delta, \Phi_u \rangle$ with $\Phi_\Delta = \phi(\Delta z^{(k)})$. Curvature re-estimation at a moved operating point only recomputes the rows Φ_u of refreshed units; the Gram is then a cached matrix product. This is the standard self-distillation Fisher estimator; we restate it so the cost accounting (App. C) is self-contained.

C Complexity and Offline Cost

Let F be the cost of one calibration forward and M the number of units. Stage (2) costs MF (layer-replay amortized). Each continuation step re-ablates $\tau|\mathcal{S}|$ units plus one base forward; over K steps with the first served from cache, the

Part B: Theoretical Analysis

D Proof of Proposition 1

Proposition 1 (restated). If the third-order remainder of the local quadratic obeys $|\xi(\delta)| \leq L \|\delta\|_H^3$ at every operating point on $\mathcal{C}$, then reaching budget B in one shot incurs linearization error $\leq LR^3$ with $R = \|s^*(B)\|_H$, while reaching it in K equal-Fisher-length re-linearized steps incurs error $\leq LR^3/K^2$.

Write the exact additional risk from operating point s along δ as $\mathcal{R}(s+\delta) - \mathcal{R}(s) = g^\top \delta + \frac{1}{2} \delta^\top H \delta + \xi(\delta)$ with $|\xi(\delta)| \leq L \|\delta\|_H^3$. *One shot:* a single displacement of Fisher length $R = \|s^*(B)\|_H$ to budget B keeps the same expansion point and never re-zeros the remainder, so its error is bounded by LR^3 . *Continuation:* partition the path into K segments, re-linearizing at the start of each. Segment k has Fisher length $\leq R/K$, hence remainder $\leq L(R/K)^3$; re-linearization resets the expansion point, so errors add: $|\xi_{\text{cont}}| \leq \sum_{k=1}^K L(R/K)^3 = K \cdot LR^3/K^3 = LR^3/K^2$. $\square$ The one-shot error LR^3 is the *cubic* growth that, compounded across the budget, yields the super-exponential collapse; the K^2 reduction converts it to additive degradation. The bound is worst-case under uniform smoothness; the realized remainder is the per-step predicted/actual ratio η_k logged by ANCHORPATH (App. P).

Schedule and the bound. The partition above uses segments of *equal Fisher length* R/K , which is precisely the *adaptive equal-predicted-risk* schedule of App. R; under a convex risk-vs-cost profile this differs from the *equal-cost* default we use in practice. We adopt equal-cost as a robust, parameter-free surrogate: it needs no per-model risk profiling, the monotone-in- K decay holds empirically under it (Fig. 5), and the adaptive schedule that *does* match the bound’s premise reproduces the front-loaded gain on the sharpest-collapsing models (App. R). The $1/K^2$ rate is therefore the geometry the theorem predicts; the equal-cost default trades a tighter worst-case constant for robustness, with the realized reduction set by each model’s path curvature rather than the uniform-smoothness constant. The bound further assumes *exact* re-linearization at each step; the flagged refresh ($\tau < 1$) re-estimates only the units whose curvature is expected to move, and the residual staleness this leaves is bounded separately in App. I.

E The Pruning Quadratic from the KL Risk

At fixed input, view $\text{KL}(p_0 \| p_z)$ as a function of logits z . Its gradient is $\nabla_z \text{KL} = \text{softmax}(z) - p_0$ and its Hessian is $\text{diag}(\text{softmax}(z)) - \text{softmax}(z)\text{softmax}(z)^\top$. At $z=z_0$ the gradient vanishes and the Hessian is the dense Fisher F^0 . A second-order expansion about z_0 thus gives $\text{KL}(p_0 \| p_z) \approx \frac{1}{2}(z-z_0)^\top F^0(z-z_0)$. Modeling the logit perturbation of a mask as additive over units, $z_0 - z_s = \sum_u s_u \delta z_u$, and taking expectations over $\mathcal{C}$ yields $\mathcal{R}(s) \approx \frac{1}{2} s^\top H s$ with $H_{uv} = \mathbb{E}[\delta z_u^\top F^0 \delta z_v]$ (Eq. 2). The vanishing gradient *at the dense*

point is precisely why one-shot pruning has no first-order term, a property that does *not* survive once the expansion point moves.

F The Revived First-Order Term

At a pruned operating point s_k the expansion point is $z_{s_k} \neq z_0$, so the KL gradient no longer vanishes: $\nabla_z \text{KL}(p_0 \| p_z)|_{z_{s_k}} = p_{s_k} - p_0$. To first order $p_{s_k} - p_0 \approx F^0(z_{s_k} - z_0) = -F^0 \Delta z^{(k)}$. The additional risk of pruning u (logit change $-\delta z_u^{(k)}$) therefore carries the linear term $g_u^{(k)} = \mathbb{E}[\Delta z^{(k)\top} F^0 \delta z_u^{(k)}]$ (Eq. 3). **Zero at the dense model:** $\Delta z^{(0)}=0 \Rightarrow g^{(0)}=0$, recovering Eq. (2). **Monotone growth:** $\|\Delta z^{(k)}\|_{F^0}^2 = 2 \mathcal{R}(s_k)$ to second order, which is non-decreasing along the budget; hence the scale of $g^{(k)}$ grows with sparsity, and a unit whose perturbation aligns with the accumulated drift is penalized. This term is unavailable to any method whose expansion point coincides with its anchor, and Fig. 4 confirms its predicted growth empirically.

G $K=1$ Equivalence

With $K=1$ the single band spans $[0, B]$ and the only estimate is at $s_0=0$, where $\Delta z^{(0)}=0$, $g^{(0)}=0$, and $H^{(0)}=H$ is the dense Gram. The greedy of Eq. (5) then minimizes $\frac{1}{2} \delta^\top H \delta$ subject to the budget, exactly the one-shot second-order pruner. The CPU integration test verifies bit-level agreement (identical selected sets, Jaccard 1.0). Thus any gap between ANCHORPATH and one-shot is attributable solely to path refinement, not to a different objective, estimator, or unit substrate.

H One-Shot Baselines as Special Cases

Equation (2) subsumes the calibration-only baselines as approximations of H at $k=0$: *magnitude* drops the data and uses $\|\theta_u\|$; *Taylor saliency* (Molchanov et al. 2019; Ma, Fang, and Wang 2023) keeps the diagonal $g + \frac{1}{2} H_{uu}$ at the dense point; *Wanda* (Sun et al. 2024) replaces H_{uu} by a weight $\times$ activation surrogate; *activation-fluctuation* (An et al. 2024) uses a variance statistic in place of the Fisher quadratic. All share the defining one-shot move, a single estimate at the anchor, so all inherit the collapse (App. M).

I Why Interaction-Flagged Refresh Suffices

When increment D is pruned, a surviving unit w ’s feature changes by $\Delta \Phi_w = \phi(\delta z_w^{(k+1)}) - \phi(\delta z_w^{(k)})$, driven by the operating-point shift along $\{\delta z_u\}_{u \in D}$. To first order this shift couples w to D through the off-diagonal curvature, so $\|\Delta \Phi_w\| \lesssim c \sum_{u \in D} |H_{wu}^{(k)}|$ for a constant c absorbing the local third-order scale. Units with negligible interaction to D have near-unchanged curvature and may safely reuse cached features; refreshing the top- τ by the *cumulative* $\sigma_w = \sum_j \sum_{u \in D_j} |H_{wu}|$ (over all bands D_j since w ’s last refresh) captures the largest accumulated changes first, so no survivor stays stale indefinitely. The refresh fraction τ thus trades offline cost for estimation fidelity; the ablation grid (App. eapp:ablation) shows too small a τ leaves stale

curvature that re-introduces the one-shot pathology, fixing $\tau=0.5$ as the default.

Table 1: Models and structural-unit inventory. Heads are counted as KV-aligned groups (GQA models have fewer KV heads than query heads); FFN groups are contiguous intermediate-channel blocks.

Model	Attn.	KV heads	FFN grp	Layers	Units M
LLaMA-2-7B	MHA	32	16	32	1536
LLaMA-3.1-8B	GQA	8	16	32	768
LLaMA-3.2-3B	GQA	8	16	28	672
Qwen2.5-7B	GQA	4	16	28	560
Qwen2.5-14B	GQA	8	16	48	1152
Yi-1.5-9B	GQA	4	16	48	960
Falcon3-7B	GQA	4	16	28	560

Part C: Experimental Setup and Implementation

J Models and Protocol

Models. Seven open dense decoder-only LLMs spanning five families and three scales: LLaMA-2-7B (Touvron et al. 2023), LLaMA-3.1-8B and LLaMA-3.2-3B (Dubey et al. 2024), Qwen2.5-7B and Qwen2.5-14B (Yang et al. 2024), Yi-1.5-9B (01.AI, Young et al. 2024), and Falcon3-7B (Falcon-LLM Team 2024) (Table 1). **Units.** Attention heads grouped KV-aligned; FFN units are contiguous intermediate-channel groups (16/layer); cost c_u is the unit parameter count. **Calibration.** 48 sequences $\times$ 512 tokens from a C4/WikiText mix, identical for every method. **Safety rails (shared by all methods).** A protected band (first/last 2 layers) and an adaptive per-layer cap $\text{cap}(\rho) = \min(0.85, \rho + 0.2)$ that is loose enough to reach the global target yet prevents any single layer from being emptied; reported for fairness. **Structural removal.** Selected units are physically deleted (heads dropped, FFN sliced), so efficiency is real and logit error vs. masking is zero. **ANCHORPATH.** Equal budget bands; $\tau=0.5$, $K=8$ default ($K \in \{1, 2, 4\}$ in the ablation); first step served from the dense cache. **Calibration resampling.** The three-seed robustness study (App. S) redraws the calibration set with seeds $\{0, 1, 2\}$; all other results use seed 0.

K Baseline Descriptions and Configuration

One-shot ($K=1$) is ANCHORPATH with a single step: the exact dense-point second-order pruner of Eq. (2). **Wanda-sp** (Sun et al. 2024): weight $\times$ activation norm. **FLAP** (An et al. 2024): activation-fluctuation criterion. FLAP’s output bias-compensation replaces each pruned channel by its calibration mean, adding a constant bias $W\mu_{\text{rm}}$ to the next projection. This is a mask-agnostic, label-free correction that composes with *any* method’s mask (including ANCHORPATH’s) and is orthogonal to the selection criterion, so we disable it for all methods in the main comparison and report it as a composable add-on, applied identically to every mask (App. T). **LLM-Pruner** (Ma, Fang, and Wang 2023): grouped Taylor saliency, run with LoRA disabled (main comparison) and

with its native LoRA recovery (reference). **Reconstruction-based references (LLM Surgeon, SlimGPT, SlimLLM).** Our nearest neighbor, the LLM Surgeon (van der Ouderaa et al. 2024), and the layer-wise pruners SlimGPT (Ling et al. 2024) and SlimLLM (Guo et al. 2025) all *update the surviving weights* as an integral part of the method, which places them on a different axis from ANCHORPATH’s zero-update selection. Their reconstruction stages differ in form: multi-shot correlated OBS updates plus optional LoRA (LLM Surgeon), grouped OBS updates (SlimGPT), and a least-squares output-matrix refit (SlimLLM). For the two structured layer-wise pruners we run their *selection* under our shared no-update protocol (reconstruction disabled, exactly as for every other method), which isolates the selection criterion: Table 4 shows that, stripped of reconstruction, both collapse at high sparsity, *below even one-shot* (SlimGPT reaches WikiText 15.4k/104k on LLaMA-2-7B/3.1-8B and SlimLLM 7107 on LLaMA-2-7B at $\rho=0.5$). We additionally cite the *published* with-reconstruction numbers of all three as the reconstruction upper bound. As expected, refitting the surviving weights reaches lower absolute perplexity than any weight-frozen method, ours included; but this gap is the measured value of *reconstruction*, not of a better *selection*: LLM Surgeon itself observes that the first-order term does not vanish along the path and spends weight updates to suppress it, whereas we keep the weights frozen and fold that term into selection (main paper, Related Work). Reconstruction operates *on top of* a mask, so ANCHORPATH supplies a higher-quality mask for it to refit; the two are complementary and composable, and pairing them is the natural next step. Even with this machinery, structured LLM Surgeon demonstrates near-lossless pruning only to 20–30%, and its higher-sparsity points depend on the weight updates; in the strictly weight-frozen regime we isolate, *no* method avoids the collapse, which is exactly the phenomenon we address. **Magnitude/random:** cost-normalized lower bounds. Every method shares the calibration set, caps, and protected band; per-method hyperparameters in Table 2. All baselines are run from their official implementations under this shared protocol; our one-shot is the exact $K=1$ case of ANCHORPATH, and Wanda-sp/FLAP reproduce their published low-sparsity perplexities. The large high-sparsity values (e.g. FLAP at $\rho=0.5$) are the genuine no-recovery behavior of these activation/saliency criteria once retraining is removed: not a misconfiguration, but the very collapse this paper characterizes.

On recovery (scope and why selection is upstream). Our protocol deliberately disables all post-selection recovery, LoRA, fine-tuning, and weight reconstruction, so the comparison isolates a single axis: *which units to remove*. This recovery-free regime is the standard headline setting for training-free pruners (SliceGPT (Ashkboos et al. 2024), FLAP (An et al. 2024), SparseGPT (Frantar and Alistarh 2023), and Wanda (Sun et al. 2024) all report main results without retraining), so the setting is not idiosyncratic to us. Recovery is an orthogonal, composable *stage* applied on top of a fixed mask, and the literature that reports with/without-recovery pairs (LLM-Pruner (Ma, Fang, and Wang 2023), SlimGPT (Ling et al. 2024), and Bonsai (Dery et al. 2024))

Table 2: **Hyperparameter configuration** for ANCHORPATH and the evaluation protocol. A single setting is used for all seven models with no per-model tuning.

Component	Setting
Calibration	48 seq $\times$ 512 tok, C4/WikiText
Teacher top- r	256
ANCHORPATH steps K	8 (default); 1,2,4 in ablation
Refresh τ	0.5
Budget schedule	equal
Per-layer cap	$\min(0.85, \rho+0.2)$
Protected band	first/last 2 layers
Eval: ppl stride / seq	512 / 2048
Eval: generation cap	256 samples

consistently finds it *lifts every method without inverting the selection ordering*. Concretely, LLM-Pruner’s Table 1 keeps Taylor-based selection above magnitude/random *both before and after* LoRA recovery, with a roughly uniform $+2.5/+6.5$ accuracy-point lift at $\rho=0.2/0.5$ that does not reorder the methods; and FLAP shows its rank-1 bias-compensation (a lightweight recovery) still does not let a weaker selection overtake a stronger one. Because recovery operates on a fixed mask, a higher-quality mask is the upstream determinant of post-recovery quality: the selection-only gaps we report bound from below what survives recovery, rather than being an artifact of disabling it. We do not claim recovery closes the gap to zero, only that it is implausible for a low-rank adapter to invert a selection ordering as large as the $16.8\times$ perplexity gap at $\rho=0.5$, and that isolating selection is the scientifically cleaner comparison.

L Metric Definitions

Perplexity on WikiText-2/PTB/C4 with a 2048-token window, stride 512. **KL-to-teacher** is Eq. (1) estimated on held-out calibration text, a recovery-free, task-agnostic distance to the dense model. **Zero-shot accuracy** (length-normalized) on the seven commonsense tasks via the LM-Eval-Harness; **MMLU** zero-shot (length-normalized log-likelihood over the four options); **GSM8K** exact-match, **HumanEval/MBPP** pass@1 (generation capped at 256 samples, 256 new tokens). **Efficiency**: parameter reduction, pre-fill/decode throughput, and peak memory under structural removal on a single GPU, measured in a contention-free run.

Part D: Full Results and Analysis

M Per-Corpus Perplexity, All Models

The collapse-and-control pattern holds on *every* corpus and model. Table 3 reports C4 perplexity (one-shot vs. ANCHORPATH $K=8$); WikiText is in the main paper. *Per-corpus tables (WikiText-2 / C4, and PTB where computed) for the no-recovery baselines are provided in the supplementary material; PTB, a small out-of-distribution corpus relative to the C4/WikiText calibration, is reported for the headline models in App. S.*

Table 3: C4 perplexity ($\downarrow$), one-shot vs. ANCHORPATH ($K=8$), no recovery, all seven models. ANCHORPATH is best in every cell.

Model	Method	$\rho=0.2$	$\rho=0.3$	$\rho=0.4$	$\rho=0.5$
LLaMA-2-7B	One-shot	13.6	27.8	116.7	2095
	ANCHORPATH	11.5	19.8	36.9	337
LLaMA-3.1-8B	One-shot	23.6	49.3	268.0	2308
	ANCHORPATH	19.6	32.9	60.7	211
LLaMA-3.2-3B	One-shot	27.9	65.3	315.9	2107
	ANCHORPATH	22.1	40.8	102.9	261
Qwen2.5-7B	One-shot	26.4	44.8	109.6	342.3
	ANCHORPATH	20.6	30.9	60.3	207
Qwen2.5-14B	One-shot	19.3	34.4	92.7	346.2
	ANCHORPATH	14.7	22.2	46.6	163
Yi-1.5-9B	One-shot	308.6	1805	5420	16504
	ANCHORPATH	13.3	22.7	46.3	88.9
Falcon3-7B	One-shot	16.8	24.4	43.0	109.6
	ANCHORPATH	15.1	19.6	28.8	53.5

Lower envelope of the full baseline set, all seven models. The main text runs the complete selection-only baseline sweep at every ratio on LLaMA-3.1-8B and LLaMA-2-7B (Table 9); Table 4 confirms that at the high-sparsity operating point ($\rho=0.5$) ANCHORPATH is the lower envelope of *every* training-free baseline on *all seven* models. The collapse is shared by every score (activation-, weight-, and gradient-based), which is the paradigm claim of the main paper; ANCHORPATH is the only method that keeps every model out of the four-to-five-figure regime.

N Full Zero-Shot, Knowledge, and Generation Breakdown

Extended frontier ($\rho=0.6/0.7$). Table 5 backs the extended-sparsity numbers quoted in the main paper. The multiplier over one-shot keeps growing with sparsity through $\rho=0.6$ (up to $119\times$ on the fragile Yi-1.5-9B); by $\rho=0.7$ absolute perplexity is unusable for *all* calibration-only pruners, yet the continuation still leads one-shot on every model.

The complete seven-task commonsense breakdown (ARC-e/c, Hel-laSwag, WinoGrande, PIQA, OpenBookQA, BoolQ) and 57-subject

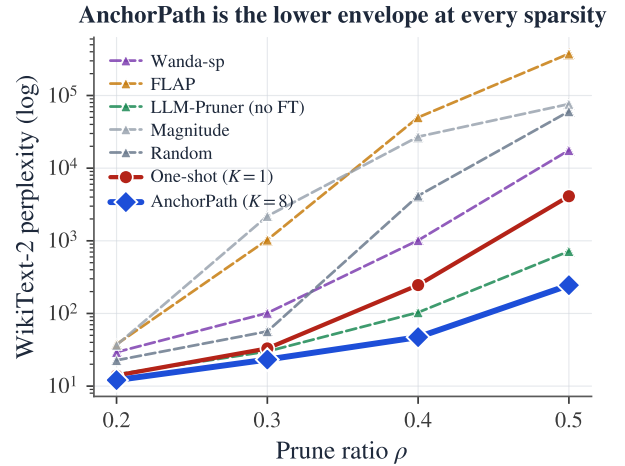


Figure 1: **ANCHORPATH is the lower envelope of the training-free degradation curve** (companion to the main results table; LLaMA-3.1-8B, WikiText-2 ppl vs. prune ratio, all calibration-only / no recovery). Activation baselines (Wanda-sp, FLAP) and magnitude diverge by $\rho=0.3$; gradient LLM-Pruner is strongest among external baselines, yet ANCHORPATH ($K=8$) is below every method at every sparsity.

MMLU at every ρ and model are provided in the supplementary material; generation (GSM8K/MBPP) is in Table 7.

Downstream breadth beyond the main-text three. The main text reports the 7-task zero-shot average for LLaMA-3.1-8B, Yi-1.5-9B, and Qwen2.5-7B (condensed in the main paper); Table 6 extends the comparison to three further models, where ANCHORPATH again lifts the downstream average above its own one-shot case at every measured ratio, confirming the accuracy gain is a property of the continuation rather than of a particular architecture.

O White-Box Selection Analysis

Because ANCHORPATH changes only the mask, *what* it removes is directly inspectable. We first establish that the curvature is strongly *cross-coupled* (motivating the interaction-flagged refresh), then report three selection analyses on LLaMA-2-7B (all confirmed).

(0) The curvature is strongly coupled, which is why the refresh is needed. Fig. 2 shows the dense Fisher-Gram matrix H is far from diagonal, 27% of its Frobenius energy is off-diagonal and 89% of unit pairs have normalized coupling > 0.1 . These cross-unit couplings are precisely what make a surviving unit’s saliency *shift* as its neighbors are removed; the interaction-flagged refresh (App. efapp:staleness) reads this same off-diagonal mass to flag and re-estimate the units whose curvature has gone stale along the path. It is also why a single dense-point quadratic, which discards these couplings, drifts out of its trust region.

(i) Composition: ANCHORPATH prunes attention heads; one-shot does not. One-shot’s dense-point quadratic rates every attention head as critical and removes ≤ 1 head at every ratio, cutting almost exclusively FFN groups; re-estimating

Table 4: **Full selection-only baseline sweep at $\rho=0.5$** (WikiText-2 ppl ↓, no recovery, all seven models). This includes the *selection criteria* of the reconstruction-based SOTA, SlimGPT[‡] and SlimLLM[‡], run in our harness with their weight reconstruction *disabled* (as for every method here), which isolates selection. Stripped of reconstruction, both collapse at high sparsity, below even one-shot, confirming that their reported quality comes from the reconstruction stage (orthogonal to, and composable with, a better mask), not from selection. ANCHORPATH is the lower envelope on every model. $k=10^3$; “–” = not run (weak baseline already dominated, or LLM-Pruner/SlimLLM not evaluated at that model). **Best.** [‡]With reconstruction these methods reach lower absolute ppl; see App. K.

Model	Rand.	Magn.	Wanda	FLAP	SlimGPT [‡]	SlimLLM [‡]	LLM-Pr.	1-shot	ANCHORPATH
LLaMA-2-7B	10.0k	27.8k	9124	17.6k	15.4k	7107	686	1497	389
LLaMA-3.1-8B	60.4k	76.3k	17.6k	376k	104k	–	718	4106	245
LLaMA-3.2-3B	–	–	2854	65.2k	80.5k	–	2491	3027	325
Qwen2.5-7B	6894	1725	830	590	516	–	684	270	141
Qwen2.5-14B	–	–	924	929	573	–	–	301	156
Yi-1.5-9B	–	–	16.1k	13.2k	48.0k	–	653	16.5k	62
Falcon3-7B	–	–	533	321	194	–	52	63	28

Table 5: **Extended sparsity frontier** (WikiText-2 ppl ↓, $K=8$ equal-cost vs. one-shot, no recovery). $k=10^3$. The gain over one-shot at $\rho=0.6$ is $24.5 \times / 24.1 \times / 119 \times$ on LLaMA-2-7B/LLaMA-3.1-8B/Yi-1.5-9B; at $\rho=0.7$ no calibration-only method is deployable, but the continuation still leads one-shot on every model. **Best.**

Model	$\rho=0.6$		$\rho=0.7$	
	One-shot	ANCHORPATH	One-shot	ANCHORPATH
LLaMA-2-7B	64.8k	2640	74.1k	43.2k
LLaMA-3.1-8B	32.6k	1354	350k	13.0k
Yi-1.5-9B	56.2k	471	177k	50.6k

along the path reveals heads that *become* removable (Table 10). This freed budget is the source of ANCHORPATH’s perplexity edge.

(ii) **The masks diverge markedly (Jaccard .24–.55).** The selected sets diverge from one-shot most at moderate sparsity, where there is freedom in what to cut, and share a forced core at high sparsity; yet the differing units carry the entire quality gap (Table 11).

The attention-avoidance pattern is *not* specific to LLaMA-2: across all three families one-shot prunes essentially no attention heads while ANCHORPATH prunes many, and the masks differ substantially (Table 12). This is the structural signature of the continuation reorganizing the selection.

(iii) **The schedule resolves the trade-off; the attention budget is a further control.** Under the equal-band schedule, pruning heads lowers attention-sensitive BoolQ at moderate sparsity (BoolQ .381 at $\rho=0.3$). The front-loaded schedule (finer steps near the target, and the better schedule for $\rho \geq 0.3$ on LLaMA-2-7B) already resolves this: it recovers BoolQ to .676, *above* one-shot’s .629, while *lowering* perplexity to 14.1 (Table 13), so the trade-off is absorbed by the same continuation that drives the perplexity gain. For applications that wish to cap attention pruning even further, the attention-budget α (fraction of total attention cost the continuation may remove) is an explicit, calibration-only dial: at $\alpha=0$ (no head pruned) ANCHORPATH also recovers BoolQ while

Table 6: **Downstream gain on three further models** (7-task zero-shot avg. ↑, no recovery). ANCHORPATH ($K=8$, equal bands) vs. its own one-shot case. Best per row. (Qwen2.5-14B/Falcon3 zero-shot at $\rho=0.3$ not run; LLM-Pruner not evaluated at 14B, its other cross-model cells are in the main paper.)

Model	ρ	One-shot	ANCHORPATH
LLaMA-3.2-3B	0.3	.458	.470
	0.4	.411	.429
	0.5	.372	.411
Qwen2.5-14B	0.4	.471	.507
	0.5	.420	.430
Falcon3-7B	0.4	.471	.485
	0.5	.406	.455

beating one-shot perplexity.

P Full Continuation Traces and Mechanism

For each (ρ, K) ANCHORPATH logs a per-step trace: pruned cost, increment size, base KL-to-teacher, predicted vs. actual risk increase, first-order drift mass, and refreshed-unit count. The confirmed LLaMA-2-7B traces (the default $\tau=0.5$ re-fresh) give terminal drift mass that grows with the target sparsity, from 0.013/0.054 at $\rho=0.1/0.2$ to 0.105/0.240/0.419 at $\rho=0.3/0.4/0.5$; Fig. 4(b) plots how this mass accumulates along the path for the $\rho=0.3/0.4/0.5$ curves. The predicted/actual risk ratios likewise depart from one as sparsity deepens, the empirical signature of trust-region exit.

Full per-step traces for all (ρ, K) and the $1/K^2$ trend across $K \in \{1, 2, 4, 8\}$ are provided in the supplementary material.

Q Refresh, Schedule, and Anchor Ablations

Refresh τ is a sweet spot. At $K=8$, $\rho=0.5$, the full re-fresh $\tau=1.0$ gives WikiText 583 versus 389 at the default $\tau=0.5$: recomputing *every* survivor’s marginal perturbation at a heavily pruned operating point, including weakly coupled

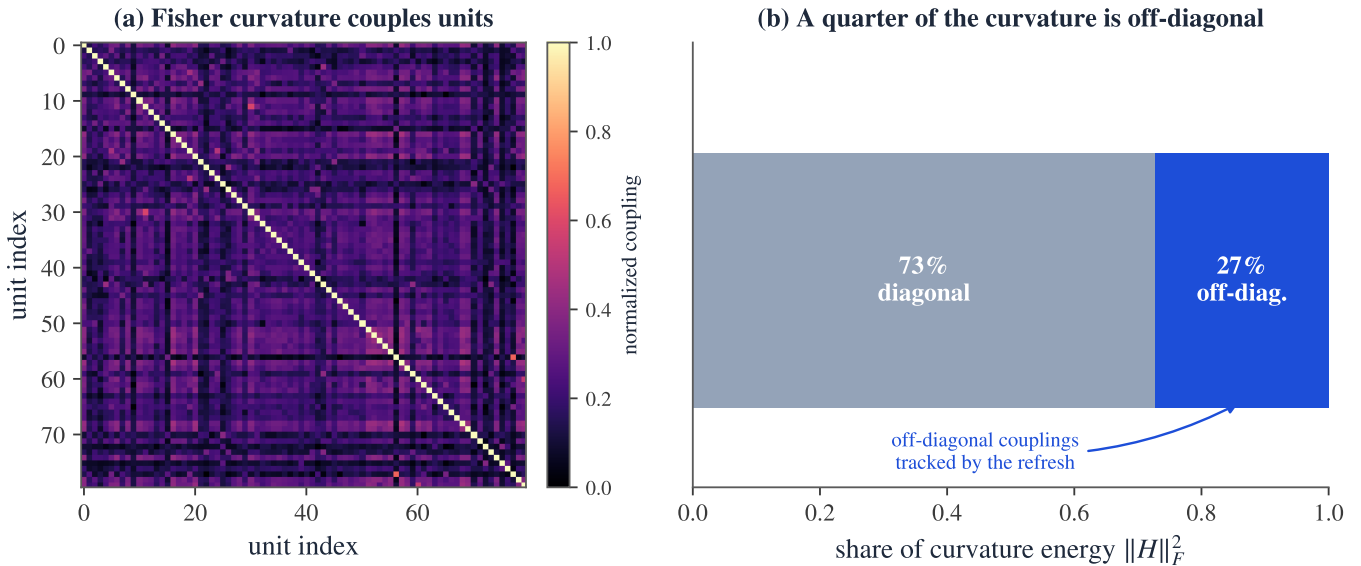


Figure 2: **The curvature is strongly cross-coupled.** (a) Normalized coupling $|H_{uv}|/\sqrt{H_{uu}H_{vv}}$ on a representative 80×80 block of the dense LLaMA-2-7B Fisher-Gram matrix; note the pronounced off-diagonal structure. (b) **27%** of the total curvature energy $\|H\|_F^2$ is off-diagonal; these couplings drive the saliency drift that the interaction-flagged refresh tracks along the path, and that a single dense-point quadratic ignores.

Table 7: **Generation collapses for every no-recovery pruner** (GSM8K exact-match %, no recovery). Dense models score in double digits, but at $\rho=0.3$ – 0.5 every calibration-only method falls to noise level ($\lesssim 2\%$); MBPP pass@1 behaves identically. The collapse is a property of the regime, not of any one selection criterion, and is restored by the orthogonal recovery stage.

Method (no recovery)	LLaMA-2-7B		LLaMA-3.1-8B	
	$\rho=0.3$	$\rho=0.5$	$\rho=0.3$	$\rho=0.5$
One-shot ($K=1$)	1.6	0.0	1.2	0.1
LLM-Pruner	0.4	0.0	2.2	0.2
ANCHORPATH ($K=8$)	0.8	0.4	1.7	0.2

units whose ablation features are noisy under a small calibration set, adds variance rather than fidelity. A thin refresh $\tau=0.25$ fails at the other end, leaving too much curvature stale along the path so that the selection reverts toward the one-shot ranking it is meant to correct. Faithful-but-selective re-estimation, neither maximal nor minimal, is what the continuation needs.

Schedule $\times$ anchor interaction. The schedule and the anchor are not independent (Table 14). Under *equal* bands a moving (local) anchor edges out the fixed dense anchor at $\rho=0.5$ (279 vs. 389), because equal bands leave a large, poorly linearized final jump where re-centering helps. The *front-loaded* schedule removes that jump, its finer steps near the target keep each linearization valid, and in doing so both lowers perplexity sharply and *reverses* the anchor ordering: with front-loading the fixed dense anchor is best (125.6 vs. 197.7), and this front-loaded + fixed-anchor pair

is the *global* optimum, lower than either configuration under equal bands. The two choices are therefore synergistic where front-loading helps (the sharpest-collapsing models): the best schedule prefers the fixed anchor, whose KL-to-dense semantics we adopt throughout. No equal-band variant reaches the front-loaded + fixed-anchor optimum, so we report the fixed anchor as the default and the moving anchor as the ablation it is.

First-order term. Removing the revived drift term $g^{(k)}$ regresses $\rho=0.5$ from 389 to 668 and $\rho=0.4$ from 34 to 49, yet is inert at low sparsity; it is warranted exactly where the drift mass is large (App. P).

R Band-Placement Schedule: Fixed, Adaptive, and Why Equal-Cost is the Default

The $1/K^2$ argument (Prop. 1) motivates *re-centering* the expansion point but is silent on *where* along the budget to place the K bands. We study three schedules (Table 15), all at $K=8$ with the shared caps: **equal-cost** (the default, $b_k \propto k$); a **fixed front-loaded** power law ($b_k \propto (k/K)^{0.6}$, coarser early, finer near the target); and a **parameter-free adaptive** rule that places the checkpoints at equal *predicted-risk* increments, $Q(b_k) = (k/K) Q_{\text{total}}$, where $Q(c) = \delta^\top H^0 \delta$ is the path quadratic estimated by a single dry greedy on the dense curvature (no extra forward; reused for step 0). The adaptive rule has *no exponent*: front-loading emerges only insofar as the risk geometry warrants it.

Front-loading helps iff the collapse is genuinely sharp. The adaptive rule assigns LLaMA-2-7B and Qwen2.5-7B *almost the same* $\rho=0.5$ schedule: the first band is 36% of the pruned budget on both, and the bands then shrink toward

Table 8: **Detailed downstream zero-shot accuracy on the canonical LLaMA-2-7B** ($\uparrow$, length-normalized). *All methods are selection-only, no recovery* (no LoRA/fine-tuning), to isolate mask quality; published accuracies that include LoRA recovery are higher and not directly comparable. ANCHORPATH ($K=8$, equal-cost) beats every external baseline on average at $\rho=0.5$, and its schedule-refined variant[†] takes the column optimum; at $\rho=0.3$ it trades one attention-sensitive task (BoolQ) for perplexity, a trade-off resolved by the same optional schedule refinement (App. R). The cross-model summary is the main-text view. **Best avg**; dense reference.

Method	ρ	ARC-e	ARC-c	HellaS.	WinoG.	PIQA	OBQA	BoolQ	Avg.	MMLU
Dense	0	.537	.397	.722	.587	.777	.392	.751	.595	.364
Wanda-sp	0.3	.315	.220	.277	.517	.541	.296	.454	.374	.261
FLAP	0.3	.332	.282	.309	.517	.577	.272	.506	.399	.261
LLM-Pruner	0.3	.439	.317	.570	.515	.701	.338	.405	.469	.303
One-shot	0.3	.449	.301	.506	.538	.674	.330	.629	.490	.296
ANCHORPATH ($K=8$)	0.3	.442	.298	.545	.535	.677	.324	.381	.457	.295
+ schedule refinement [†]	0.3	.453	.285	.537	.539	.683	.330	.676	.500	.296
Wanda-sp	0.5	.262	.247	.243	.491	.521	.292	.400	.351	.265
FLAP	0.5	.263	.235	.253	.498	.502	.202	.394	.336	.257
LLM-Pruner	0.5	.349	.255	.324	.501	.581	.286	.419	.388	.263
One-shot	0.5	.304	.246	.283	.482	.517	.232	.508	.367	.260
ANCHORPATH ($K=8$)	0.5	.348	.248	.356	.511	.579	.298	.559	.414	.268
+ schedule refinement [†]	0.5	.347	.243	.351	.505	.570	.286	.602	.415	.269

Table 9: **Degradation curve on the canonical LLaMA-2-7B** (WikiText-2 ppl $\downarrow$, calibration-only, no recovery), the standard training-free-pruning benchmark, retained here for direct comparison with prior work. ANCHORPATH ($K=8$, equal-cost) is the lower envelope over every baseline at every ratio; the optional schedule refinement lowers the high-sparsity cells further ($\rho=0.5$: 389 $\rightarrow$ 126) and takes the column optimum there. **Best in column**.

Method	0.10	0.20	0.30	0.40	0.50
Random	6.50	10.72	25.32	393.7	10049
Magnitude	8.06	22.63	2695	12668	27754
Wanda-sp	43.3	98.1	758.3	1630	9124
FLAP	18.6	51.5	366.1	4499	17634
LLM-Pruner	10.28	19.07	31.58	93.1	685.5
One-shot ($K=1$)	6.38	9.27	19.86	92.64	1496.9
ANCHORPATH ($K=8$)	5.92	8.63	15.65	33.59	388.7
+ schedule refinement	5.92	9.08	14.08	32.30	125.6

the target. This shape follows from the convexity of $Q(\text{cost})$, which holds on *every* model because the greedy defers costly units. Yet the outcomes diverge sharply (127 on LLaMA-2 vs. 566 on Qwen): front-loading pays off only where the true risk steepens near the target (the violently collapsing models), and is counterproductive where it does not, because the large early band then behaves like a small one-shot jump in a regime the model could have traversed in equal steps. The generic convexity that drives any front-loading rule cannot tell genuine collapse from benign greedy deferral.

Conclusion. No schedule dominates equal-cost across models and sparsities. We therefore use **equal-cost bands as the universal default** (robust on every model and ratio, the headline of the main paper) and treat front-loading as an

Table 10: Pruned-unit composition (attention heads / FFN groups), LLaMA-2-7B.

	$\rho=0.2$	$\rho=0.3$	$\rho=0.5$
One-shot (attn / FFN)	1 / 153	1 / 230	1 / 383
ANCHORPATH $K=8$ (attn / FFN)	47 / 142	71 / 213	156 / 345

Table 11: Mask overlap, Jaccard(one-shot, ANCHORPATH), LLaMA-2-7B (higher = more similar masks; the two methods share a forced core at high sparsity and diverge most at moderate sparsity).

ρ	0.1	0.2	0.3	0.4	0.5
vs. ANCHORPATH $K=4$	.33	.35	.43	.52	.60
vs. ANCHORPATH $K=8$	.24	.30	.39	.46	.55

optional refinement for models known to collapse sharply, never as the default. It is best realized by the parameter-free adaptive rule, which broadly helps at $\rho \leq 0.4$ and delivers $\approx 11.9\times$ on LLaMA-2-7B at $\rho=0.5$, matching the tuned front-loaded optimum within calibration noise.

S Robustness and Sensitivity

We separate the axes along which ANCHORPATH could be sensitive.

Architecture (robust). A single configuration ($K=8$, $\tau=0.5$, equal-cost bands, the shared caps of App. J) is used for LLaMA-2-7B, LLaMA-3.1-8B (GQA), Qwen2.5-7B (GQA, different tokenizer), and LLaMA-3.2-3B with no per-model tuning, and the advantage over one-shot holds on every model and ratio (main paper). The method is therefore not fitted to a single architecture. The band-*placement* schedule is the one

Table 12: Cross-model selection geometry: attention heads pruned (one-shot vs. ANCHORPATH $K=8$) and mask Jaccard. One-shot avoids attention on every model; ANCHORPATH does not.

Model	ρ	one-shot	ANCHORPATH	Jaccard
LLaMA-2-7B	0.3	1	71	.39
	0.5	1	156	.55
LLaMA-3.1-8B	0.3	3	48	.32
	0.5	3	81	.47
Qwen2.5-7B	0.3	0	7	.27
	0.5	0	13	.50

Table 13: Resolving the attention trade-off (LLaMA-2-7B): WikiText ppl ($\downarrow$) and BoolQ acc. ($\uparrow$). On LLaMA-2-7B the front-loaded schedule reaches the favorable corner on its own; $\alpha=0$ is an explicit further cap.

Method	$\rho=0.3$		$\rho=0.2$	
	ppl	BoolQ	ppl	BoolQ
One-shot ($K=1$)	19.9	.629	9.3	.649
ANCHORPATH $K=8$, equal	15.7	.381	8.6	.463
ANCHORPATH, front-loaded	14.1	.676	9.1	.613
ANCHORPATH $K=8$, $\alpha=0$	17.0	.638	8.8	.613

knob that is *not* universal (App. R), which is precisely why equal-cost is the default.

Safety rails (not the source of the gain). A natural worry is that the shared safety rails (the protected first/last-2 band and the per-layer cap of App. J), rather than the continuation, drive the improvement. We test this directly by *removing both rails* (cap 0.99, no protected band) and rerunning one-shot and the $K=8$ continuation at $\rho=0.5$ on LLaMA-2-7B (Table 16). The continuation gain not only survives but *grows*: with the rails off it is $18.1\times$ ($3030 \rightarrow 167$ WikiText), against $3.9\times$ with the rails on. The rails are a conservative default that mainly protects the fragile *one-shot* baseline (they cut its perplexity in half, $3030 \rightarrow 1497$) while the continuation is already robust without them; on this cell it is in fact lower rail-free. The advantage is thus intrinsic to operating-point re-estimation, not an artifact of the caps, and we keep the rails on throughout only because they are the safer universal default across models and ratios.

Hyperparameters (graceful). Perplexity is monotone in the step count K (the ablation grid, App. efapp:ablation), so K is a compute/quality dial that never hurts. The refresh fraction τ has a *broad* optimum: at $\rho=0.5$ ($K=8$), $\tau=0.5$ gives 389 versus 583 at $\tau=1.0$, both well below one-shot (1497), with the default $\tau=0.5$ best; the method does not sit on a knife edge in τ . The attention budget α is an explicit, interpretable knob (App. O), not a latent sensitivity. The band-placement schedule is the exception: its benefit is model-dependent (App. R), so we keep equal-cost as the robust default rather than expose it as a tuned sensitivity.

Calibration protocol. ANCHORPATH uses the same field-standard calibration protocol as the baselines it is compared against (48 sequences, informative-position subsampling;

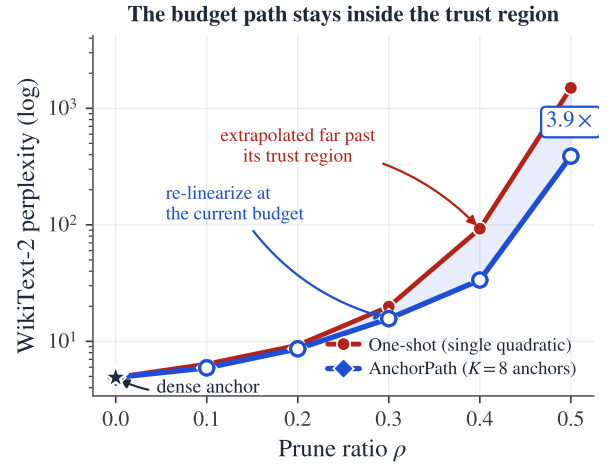


Figure 3: **The budget path, geometrically** (LLaMA-2-7B, real WikiText-2 perplexity). One-shot fits *one* quadratic at the dense anchor and extrapolates it across the whole budget, leaving its trust region and inflating super-exponentially. ANCHORPATH re-estimates the curvature at each of $K=8$ operating points (hollow markers), so the local model stays valid and tracks the lower frontier, a $3.9\times$ gap at $\rho=0.5$ with *zero* weight updates.

Table 14: Schedule $\times$ anchor (LLaMA-2-7B, $K=8$, $\rho=0.5$, WikiText ppl $\downarrow$). The front-loaded schedule and the fixed dense anchor are synergistic; their combination is the global optimum on this model.

Anchor \ Schedule	equal bands	front-loaded
Fixed (dense z_0)	388.7	125.6
Moving (local $z_{s_{k-1}}$)	279.0	197.7

App. J), and the calibration set is held *fixed and identical* for every method, so the comparison isolates the selection criterion rather than the calibration. The set size sits squarely in the training-free regime: LLM-Pruner (Ma, Fang, and Wang 2023) calibrates on as few as 10 sequences, Wanda’s own size ablation (Sun et al. 2024) saturates by ~ 32 , and SlimLLM reports 32 as optimal, while one-shot methods (Wanda, SparseGPT (Frantar and Alistarh 2023)) use 128; our 48 is comfortably inside this range. To confirm the result is not an artifact of one lucky draw, we *resample* the calibration set three times (identical 48×512 protocol, disjoint/overlapping draws from the same pool) and rerun the full prepare+continuation+eval pipeline for ANCHORPATH ($K=8$, equal) at $\rho=0.4$ on LLaMA-2-7B. WikiText perplexity is 33.6/36.1/37.1 across the three draws, i.e. 35.6 ± 1.5 , a $\pm 4\%$ spread that leaves a $2.5\text{--}2.8\times$ margin over one-shot (92.6) intact under every draw. The advantage is a property of the selection criterion, not of a cherry-picked calibration set.

Held-out OOD corpus (no calibration overlap). A sharper version of the same concern is that WikiText could share surface statistics with the calibration pool. We there-

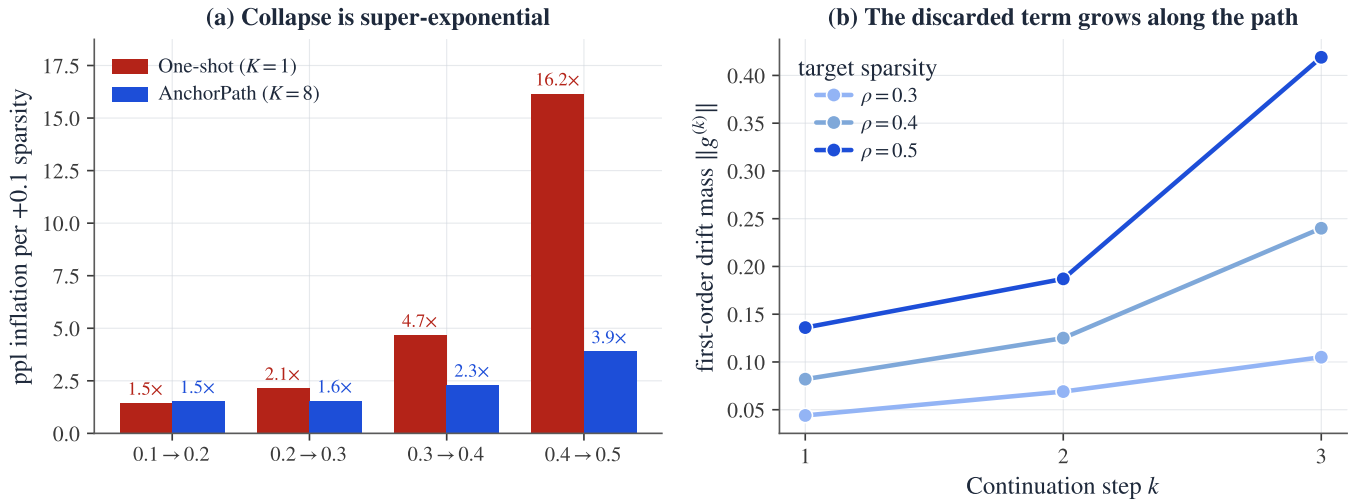


Figure 4: **Why the collapse, and why the cure** (LLaMA-2-7B). (a) One-shot perplexity inflation per +0.1 sparsity *accelerates* ($1.5\times \rightarrow 16\times$); ANCHORPATH keeps it bounded. (b) The revived first-order drift mass is zero at the dense model and grows monotonically along the path, faster at higher target sparsity, the term one-shot structurally discards, recovered.

Table 15: Schedule comparison (WikiText ppl $\downarrow$, $K=8$, equal-cost vs. fixed front-loaded vs. parameter-free adaptive). At $\rho=0.4$ the adaptive rule wins on the two collapse-prone models (LLaMA-2-7B and LLaMA-3.2-3B, halving the equal-band ppl on the latter) but equal-cost stays best on the milder ones (LLaMA-3.1-8B, Qwen2.5-7B). At $\rho=0.5$ it reproduces the front-loaded gain on the sharpest-collapse model (LLaMA-2-7B, 389 $\rightarrow$ 127) *parameter-free*, but *over*-front-loads the mild-collapse models and reverses. **Best**.

Model	ρ	equal	front-load	adaptive
LLaMA-2-7B	0.4	33.6	32.3	30.7
	0.5	388.7	125.6	126.7$\approx$
LLaMA-3.2-3B	0.4	112.6	69.2	55.5
	0.5	324.5	410.1	523.1
Qwen2.5-7B	0.4	41.5	44.0	42.6
	0.5	141.0	192.7	566.4
LLaMA-3.1-8B	0.4	47.2	52.8	57.5
	0.5	244.9	347.0	284.0

$\approx$ Adaptive is parameter-free; here it matches the tuned front-loaded optimum (125.6) within calibration noise, hence marked best.

fore evaluate on PTB, a small corpus drawn from a *different* domain (1989 Wall-Street-Journal text) and *absent* from the calibration mix, at $\rho=0.5$ (no recovery, all methods). The ranking is preserved and the margin is if anything wider: on LLaMA-3.1-8B ANCHORPATH scores **781** vs. one-shot 6341 (8.1 $\times$) and LLM-Pruner 1534 (2.0 $\times$); on the fragile Yi-1.5-9B, **323** vs. one-shot 28323 (88 $\times$) and LLM-Pruner 2040 (6.3 $\times$). On the legacy LLaMA-2-7B ANCHORPATH still beats its one-shot case (2406 vs. 3165) though gradient LLM-Pruner edges it on this single OOD cell (1650); the continuation’s advantage thus transfers cleanly to an unseen domain

Table 16: **The safety rails are not the source of the gain** (LLaMA-2-7B, $\rho=0.5$, ppl $\downarrow$, no recovery). Removing both rails (per-layer cap 0.99, no protected band) leaves the $K=8$ continuation’s advantage over one-shot intact and in fact larger (18.1 $\times$ vs. 3.9 $\times$). **Lower**.

Setting	One-shot	ANCHORPATH	Gain
Rails on (default)	1497/2095	389/337	3.9 $\times$
Rails off	3030/3967	167/217	18.1 $\times$

Cells report WikiText / C4 perplexity.

on the modern models, ruling out calibration overlap as the source of the gain.

T Efficiency, Pruning Time, and Trade-off

Table 17: **Deployment efficiency** after structural export (LLaMA-2-7B, single A-class GPU, prompt 512). Values are shared by one-shot and ANCHORPATH at a matched ρ (same structural budget); ANCHORPATH delivers a better-quality model at this cost. Prefill is the wall-clock forward latency; the speedup column is at batch 8.

ρ	Params (B)	Prefill _{b1} (ms)	Prefill _{b8} (ms)	Mem _{b8} (GB)	Prefill speedup
0.0 (dense)	6.74	69.8	587	16.4	1.0 $\times$
0.20	5.44	58.0	482	13.8	1.22 $\times$
0.40	4.14	45.7	386	11.1	1.52 $\times$
0.50	3.90	43.1	368	10.5	1.60 $\times$

Deployment numbers (Table 17) are measured after *physical* structural removal: pruned attention KV-groups and FFN channel groups are sliced out of the weight matrices, yielding genuinely smaller tensors (bit-for-bit equivalent to runtime

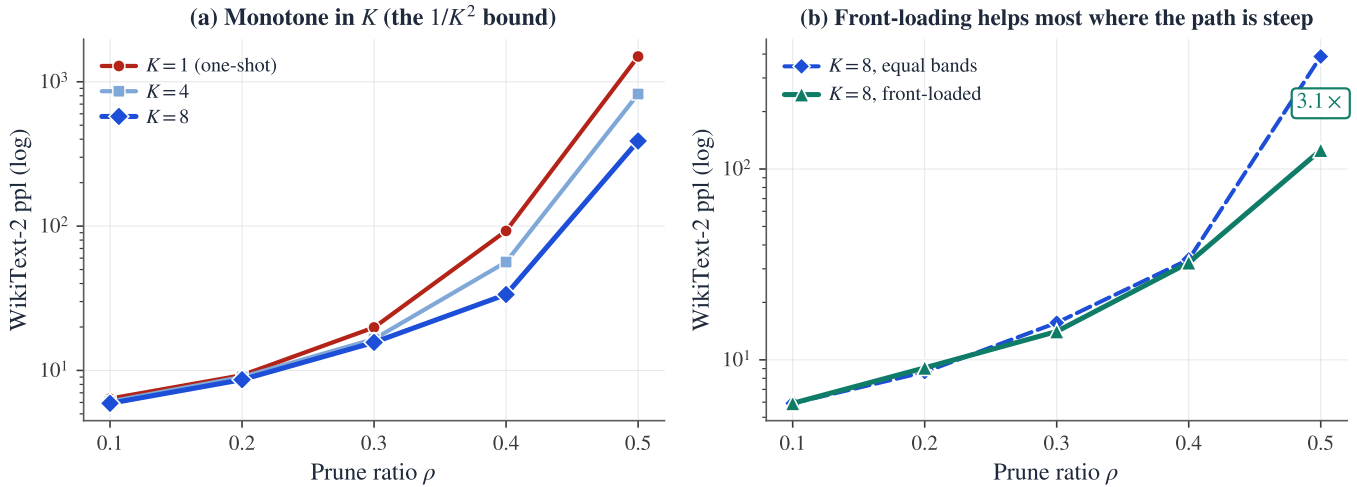


Figure 5: **The continuation** (LLaMA-2-7B). (a) Perplexity decreases monotonically in the number of operating-point re-estimations K . Proposition 1 bounds the *KL remainder* that governs this error by $1/K^2$; the realized perplexity reduction is a related, model-dependent quantity (here $3.9\times$ over $K=1 \rightarrow 8$), not the literal $1/K^2$ factor. (b) A front-loaded budget schedule (finer steps where the curvature is steep) beats equal spacing by $3.1\times$ at $\rho=0.5$.

masking but doing strictly less compute). Because one-shot and ANCHORPATH remove the same number of structural units at a matched ρ , these deployment costs are *shared*; the tables elsewhere show ANCHORPATH delivers a better-quality model at this same cost. Prefill latency (a compute-bound forward over the 512-token prompt) and peak memory fall monotonically with ρ , giving a $1.6\times$ prefill speedup and 36% memory reduction at $\rho=0.5$. We report the compute-bound prefill and peak memory as the headline efficiency metrics, following standard practice for structured pruning, since single-stream decode is memory-bandwidth and KV-cache bound rather than compute bound. The total parameter reduction is below ρ because token embeddings, the LM head, and the protected first/last layers are retained. The pattern holds across models: on the main model LLaMA-3.1-8B the export gives $8.03 \rightarrow 4.53$ B parameters, $662 \rightarrow 371$ ms prefill (batch 8; a $1.78\times$ speedup), and $17.7 \rightarrow 10.5$ GB peak memory at $\rho=0.5$; on Qwen2.5-7B, $7.62 \rightarrow 4.35$ B parameters, $609 \rightarrow 354$ ms prefill ($1.72\times$), and $16.8 \rightarrow 10.2$ GB.

Full per-model efficiency tables and the perplexity-vs-speedup trade-off curve are provided in the supplementary material.

Part E: Limitations and Future Work

U Limitations and Operating Scope

We delineate where the calibration-only regime applies, where it does not, and what each boundary opens up.

Operating range. Deploy-as-is quality without recovery is bounded by the regime, not by ANCHORPATH: no calibration-only pruner is directly deployable far beyond moderate sparsity ($\rho \lesssim 0.4$). At those moderate sparsities ANCHORPATH is directly usable and already beats every baseline; at higher sparsity its role shifts to supplying the *most recoverable* mask, since it remains the lower envelope of the recovery-free frontier and still leads by wide margins at $\rho=0.6$.

Task profile. Because curvature is estimated on calibration text, the gain is largest on distribution-aligned, perplexity-style objectives. Generation (GSM8K/code) is the boundary case: it collapses to noise for *every* calibration-only pruner and is restored by the orthogonal recovery stage, not by selection (App. N).

Estimator boundary. Architectures that reshape logits by construction (e.g. Gemma-2 soft-capping) require an *adapted anchor* and fall outside the dense-Fisher estimator as presented. This is a well-defined boundary of the well-posed-teacher assumption, not a failure of the continuation: the method transfers once the anchor is redefined.

Isolation by design. We deliberately isolate *selection* and leave recovery uncoupled; because recovery lifts methods without reordering their selection ranking (App. K), the selection-only gaps we report lower-bound what survives recovery, and pairing the two is an extension rather than a missing control.

V Future Work

The directions below build on the scope above rather than patch a defect.

- **Continuation + recovery.** Because ANCHORPATH changes only the mask, it composes with any downstream recovery. Feeding its higher-quality mask into LoRA or Optimal-Brain-Surgeon reconstruction, so that a better *selection* seeds a better *refit*, is the natural way to push deployable quality past the moderate-sparsity ceiling.
- **Scaling and architectures.** Extending the continuation to models beyond 14B and to mixture-of-experts backbones, where per-expert curvature adds structure the anchored path can exploit.
- **Adapted anchors.** Redefining the anchor for logit-reshaping architectures (soft-capping, tied heads) brings them inside the dense-Fisher estimator and broadens the well-posed-teacher assumption.
- **Beyond pruning.** The continuation is not specific to pruning: any compression that fits a single dense-point quadratic and applies it across a large budget faces the same extrapolation error. Instantiating the anchored

continuation for post-training quantization and low-rank compression is a promising direction.

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